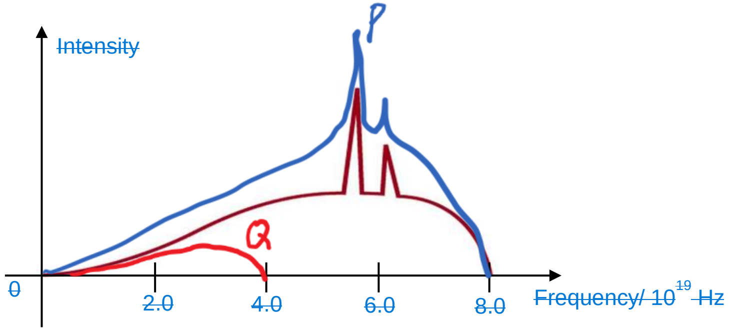
8(a) When the most energetic electrons lose all its KE at one go, it will produce a photon with the highest frequency.

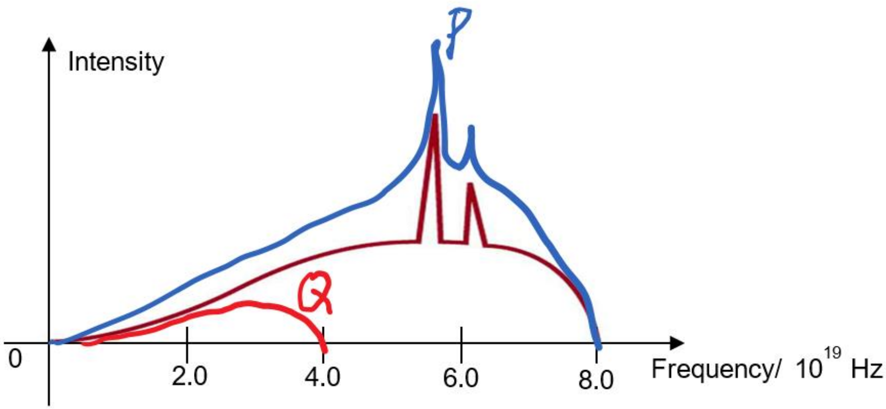
$KE_{max} = hf$

$= 6.63 \times 10^{-34} \times 8.0 \times 10^{19}$

$= 5.3 \times 10^{-14}$ J



(b) Higher filament current means more electrons incident on metal target, but KE of electrons remain unchanged. Hence graph P has higher intensity, but the maximum frequency remains the same, with the same characteristic wavelengths.



(c) Since accelerating potential is halved means that KE of electrons bombarding the metal target is halved. Hence the maximum frequency of x-ray photon is halved, ie. Graph Q must cut at max frequency = 4.0×10^{19} Hz.

The intensity of Q will be lower than the original curve.

The characteristic wavelengths would not be produced.